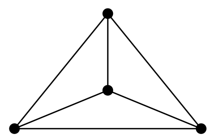
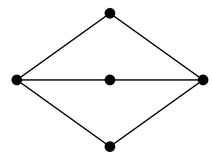
Chapter 7: **Planar Graphs**

## 7.1 Plane Graphs

A ***plane graph*** is a graph embedded in the plane such that no pair of lines intersect. The graph divides the plane up into a number of regions called ***faces***. A graph is ***planar*** if it has a planar embedding. Here are embeddings of $K_4$ and $K_{2,3}$.





For example, a plane graph has one face iff it is a forest. Further, if a graph is 2-connected, then every face is bounded by a cycle. This uses Jordán's curve theorem, which is actually more difficult to prove than it looks.

Euler's formula. *$V - E + F = 2$ for connected plane graph with $V$ vertices, $E$ edges and $F$ faces.*

Proof. By induction on $E$. If $E = V - 1$, then $G$ is a tree and we're done. Otherwise $E \geq V$ and there is a cycle containing some edge $e$. The removal of $e$ merges two faces. $\square$

It follows that:

Bound. *For plane graph, $E \leq 3V - 6$*

Proof. Let $M$ be the number of edge–face pairs. Each edge appears twice. Each face appears at least three times. So we get $2E = M \geq 3F$. That is, $F \leq 2E/3$. Plug, into formula and algebra. $\square$

As a consequence, we get two important facts: (i) that $K_5$ is not planar, and (ii) that a planar graph has minimum degree at most 5. A ***maximal planar graph*** is one where every face is a triangle; equivalently one where $E = 3V - 6$.

If a graph is bipartite, one can improve the bound, since every region has a boundary of size at least 4:

Bipartite planar. *If $G$ is bipartite, then $E \leq 2V - 4$.*

Consequence: $K_{3,3}$ is not planar.

## 7.2 Duals, Drawings, and Outerplanar Graphs

The ***dual*** of a plane graph is obtained by placing a vertex inside each face, and joining two such vertices if the faces are adjacent. Note that the dual has the same number of edges as the original. Indeed, each edge $e$ has a partner $e^*$ in the dual. If the graph is suitably connected, then the dual of the dual is the original. It can be shown that a set of edges forms a cycle in the graph if and only if the set of dual edges form a minimal cut in the dual.

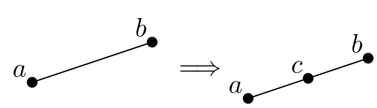
If we wrap a drawing onto a sphere, and then off again, we can move any face to be the exterior face. Every planar graph has an embedding such that all edges are line segments. We omit the proofs of the following:

Drawings. *If $G$ is 3-connected, then:*
*(a) the face boundaries are its non-separating induced cycles*
*(b) any two planar embeddings are "equivalent".*

A graph is ***outerplanar*** if it has an embedding in the plane with all the vertices on the outer face. It can be shown that neither $K_4$ nor $K_{2,3}$ is outerplanar. Also that the minimum degree of outerplanar graph is at most 2. And that its dual is the join of $K_1$ and a tree.

## 7.3 Kuratowski and Wagner

Recall that a ***subdivision*** of a graph is obtained by adding vertices of degree two into edges. Clearly, taking a subdivision of a graph preserves whether it is planar or not.



KURATOWSKI'S THEOREM. *A graph is planar if and only if it does not contain a subdivision of either $K_5$ or $K_{3,3}$.*

PROOFSKETCH. We saw earlier that $K_5$ and $K_{3,3}$ are not planar. So neither is any of their subdivisions.

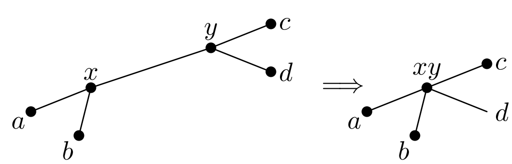
Suppose $G$ is not planar. We may assume $G$ is minimal nonplanar; that is, removal of any edge or vertex yields a planar graph. In particular it follows that $G$ is 2-connected. (In general a graph is planar if and only if all its blocks are planar.) Further, we may assume that $G$ is not the subdivision of another graph.

Now, we claim $G$ has minimum degree at least 3. Suppose vertex $v$ has degree 2 with neighbors $x$ and $y$. If $x$ and $y$ not already adjacent, then splice $v$ out—delete $v$ and add edge $xy$. Then resultant graph is such that $G$ is subdivision thereof, a contradiction of the minimality of $G$. And if $x$ and $y$ are already adjacent, then one can delete $v$, embed $G - v$ and add back in $v$ in either region where $xy$ is a boundary. Thus the claim is proven.

Now, it follows that there is an edge $e = uv$ such that $G - e$ is 2-connected. (Proof omitted.) By the assumption on $G$, the graph $G - e$ is planar.

Consider a cycle $C$ of $G - e$ containing $u$ and $v$. We would like to add edge $e$. So we may assume that there are barriers to doing this inside and outside $C$. If we make the interior of $C$ as large as possible, we can assume properties outside $C$. Then argue that no matter how place inside $C$, one obtains a $K_5$ or $K_{3,3}$ subdivision. (Details omitted.) ◊

The ***contraction*** of edge $e$ is to delete the ends of $e$ and build a new vertex adjacent to all vertices originally adjacent to one or both the ends of $e$. A graph $G$ is a ***minor*** of $H$ if $G$ can be obtained from $H$ by deleting vertices and contracting edges. It is straightforward that: If $G$ is a subdivision of $H$, then $H$ is a minor of $G$. Again, contractions preserve planarity; that is, if $G$ is planar then so is any contraction of $G$.



WAGNER'S THEOREM. *A graph is planar if and only if it contains either $K_5$ or $K_{3,3}$ as a minor.*

One direction is a consequence of Kuratowski's theorem: if $G$ is not planar it contains a subdivsion of one of the graphs and hence a minor. On the other hand, if $G$ is planar, then all its minors are planar; so it cannot contain either $K_5$ or $K_{3,3}$ as a minor.

## 7.4 Chromatic Number

We noted that by Euler's formula, every planar graph has vertex of degree at most 5. It follows that every planar graph is 6-colorable.

Every planar graph is 4-colorable. Kempe thought he proved this. But proof wrong; his approach was salvaged to prove that every planar graph is 5-colorable:

KEMPE CHAINS. *Every planar graph is* 5*-colorable.*

PROOF. The proof is by induction on the order. We noted above that every planar graph $G$ has a vertex $v$ of degree at most 5. The graph $G-v$ is 5-colorable by the induction hypothesis. One can re-insert vertex $v$, and color it, unless it has a neighbor of each color.

Consider a planar drawing of $G$ where vertex $v$ has neighbors $v_1$ through $v_5$ in cyclic order; assume each $v_i$ has color $i$. Let $H_{i,j}$ be the subgraph of $G-v$ induced by the vertices of colors $i$ or $j$. If vertices $v_1$ and $v_3$ are not connected in $H_{1,3}$, then one can interchange colors 1 and 3 in the component thereof containing $v_1$, and thus get to situation where $v$ has no neighbor of color 1; whence we are done. So we may assume that vertices $v_1$ and $v_3$ are connected in $H_{1,3}$. Similarly, we may assume that vertices $v_2$ and $v_4$ are connected in $H_{2,4}$. But these two connections contradict the planarity. ◻

The famous 4-color theorem of Appel, Haken, and Koch used computers and showed that every planar graph is 4-colorable. (Robertson, Sanders, Seymour, Thomas redid.)

The elegant result is by Thomassen:

Planar Choosable. *Every planar graph is* 5*-choosable.*